The $\mathcal{N}=2$ SYK generalization

Matrix Robin responses and superconformal reconstruction

The relative-cylinder construction in your draft [1] has an explicit extension to $\mathcal{N}=2$ SYK. The neutral ladder needs **two scalar fields with a constant matrix boundary condition**. The charged ladder has a **Dirac-field response**. Their masses stay fixed; repeated rungs change only the quadratic boundary terms. A fixed integral transform then reconstructs superconformal blocks.

This note records the resulting four-point identities and their normalization. It does *not* assert a supersymmetric interacting bulk dual. In particular, supersymmetry of the reconstructed CFT data and supersymmetry of the auxiliary bulk boundary problem are different questions.

1. Setup and the two neutral channels

Work at leading large N, at the charge-symmetric saddle, in the infrared conformal limit. The odd integer $q \geq 3$ counts fermions in the *supercharge*. Set

$$\delta = \frac{1}{q}, \qquad \nu = 1-\delta, \qquad \mu = \frac{1}{2}-\delta, \qquad L_q = 2(q-1)4^{-\nu}. \tag{1}$$

In the component conventions of Ref. [2],

$$G_\psi(t) = b_\psi \frac{\operatorname{sgn} t}{|t|^\delta}, \qquad G_b(t) = \delta b_\psi |t|^{-1-\delta}, \qquad J b_\psi^q = \frac{\tan(\pi\delta/2)}{2\pi}. \tag{2}$$

Here b labels the auxiliary boson of the SYK model; b_ψ is a propagator amplitude. The external dimensions are $\Delta_\psi = \delta/2$ and $\Delta_b = (1+\delta)/2$. Throughout, g is a *formal rung weight at fixed propagators*. Physical conformal SYK has $g=1$; no microscopic supersymmetric line parametrized by g is assumed.

The neutral pair channels $\psi\bar\psi$ and $b\bar b$ mix. With the ordered-pair integration measure, the component kernels factor as

$$\begin{aligned} \mathsf{K} &= \mathsf{C}_0\mathsf{V}, \qquad \mathsf{C}_0 = \operatorname{diag}\left(G_\psi(t_{14})G_\psi(t_{23}),\, G_b(t_{14})G_b(t_{23})\right), \\ \mathsf{V}(t) &= J(q-1)\begin{pmatrix} (q-2)G_b(t)G_\psi(t)^{q-3} & G_\psi(t)^{q-2} \\ G_\psi(t)^{q-2} & 0 \end{pmatrix}. \end{aligned} \tag{3}$$

These are the component kernels of Ref. [2], written to expose the two rails and the insertion. Let $p=0,1$ denote even or odd parity of the fermion-pair component, and put $\sigma=(-1)^p$. The boson-pair component has parity $1-p$: the off-diagonal factor G_ψ^{q-2} is odd.

Integrate over the separation of pair centers and write the relative coordinate as $x = e^s n$, $n=\pm1$. The two fixed weights are

$$a_\psi = \delta - \frac{1}{2} = -\mu, \qquad a_b = \delta + \frac{1}{2} = 1-\mu, \qquad (W_i f_i)(s,n) = e^{a_i s} f_i(e^s n). \tag{4}$$

Both pair Casimirs become $\partial_s^2 - 1/4$; a component dimension h is represented by $\omega = h - 1/2$. If $V_{ij} \propto |x|^{-v_{ij}}$, then

$$v_{11} = 2-2\delta, \qquad v_{12} = v_{21} = 1-2\delta, \qquad 1 - a_i - a_j - v_{ij} = 0.$$

This cancellation is why the boundary mixing is constant, despite the unequal external dimensions.

2. The normalized matrix Robin identity

Use global AdS_2 of radius one, with $\zeta = \cos\theta$. The two scalar masses and boundary expansions are

$$\begin{aligned} m_\psi^2 &= \mu(\mu+1), & \Phi_\psi &= A_\psi \zeta^{-\mu} + B_\psi \zeta^{1+\mu} + \cdots, \\ m_b^2 &= \mu(\mu-1), & \Phi_b &= A_b \zeta^{\mu} + B_b \zeta^{1-\mu} + \cdots. \end{aligned} \tag{5}$$

Both Φ_ψ and Φ_b are scalars; their subscripts name the pair channels, not bulk statistics. A regular scalar of index $\rho > 0$ has the response

$$R_{\rho,p}(\omega) = \frac{B}{A} = \frac{\Gamma(-\rho)}{\Gamma(\rho)} \prod_{\pm} \frac{\Gamma\left(\frac{p+1/2+\rho\pm\omega}{2}\right)}{\Gamma\left(\frac{p+1/2-\rho\pm\omega}{2}\right)}. \tag{6}$$

The formulas use unit-normalized parity harmonics $(1, \pm 1)/\sqrt{2}$ on the two strip endpoints. An individual endpoint covariance is half the corresponding mode covariance.

The first field uses alternate quantization and the second standard quantization. The boundary variables and their conjugate free sources are

$$X = \begin{pmatrix} A_\psi \\ B_b \end{pmatrix}, \qquad j_0 = \begin{pmatrix} -2\nu B_\psi \\ 2\delta A_b \end{pmatrix}, \qquad \mathsf{D}_p(\omega) = \begin{pmatrix} -[2\nu R_{\nu,p}(\omega)]^{-1} & 0 \\ 0 & R_{\delta,1-p}(\omega)/(2\delta) \end{pmatrix}. \tag{7}$$

Thus $X = \mathsf{D}_p j_0$. In particular the second entry is a *standard-quantized response*, not a second Neumann inverse.

Define

$$\mathsf{S} = \operatorname{diag}\left(1, \frac{1}{2\delta}\right), \qquad \mathsf{M}_p = \sigma L_q \begin{pmatrix} 2\mu & 2\delta \\ 2\delta & 0 \end{pmatrix}. \tag{8}$$

If $\mathsf{k}_p(h)$ is the component ladder multiplier, the entrywise identity is

$$\boxed{\widehat{\mathsf{k}}_p(h) := \mathsf{S}\mathsf{k}_p(h)\mathsf{S}^{-1} = \mathsf{D}_p(h-1/2)\mathsf{M}_p.} \tag{9}$$

It follows either from the reduced rails or from the component gamma-function kernels. It holds at generic complex h, not only at their poles. Consequently

$$\boxed{\mathsf{G}_{g,p} = (\mathsf{D}_p^{-1} - g\mathsf{M}_p)^{-1}, \qquad (I - g\widehat{\mathsf{k}}_p)^{-1}\widehat{\mathsf{k}}_p = \mathsf{G}_{g,p}\mathsf{M}_p.} \tag{10}$$

A quadratic boundary term $-\frac{g}{2}\int \mathrm{d}s\, X^T \mathsf{M}_p X$ gives $j = j_0 - g\mathsf{M}_p X$. With no external source, its boundary conditions are

$$-2\nu B_\psi = g\sigma L_q(2\mu A_\psi + 2\delta B_b), \qquad A_b = g\sigma L_q A_\psi. \tag{11}$$

At $g = 0$ these are $B_\psi = 0$ and $A_b = 0$. All ladder insertions remain on the boundary; the bulk equations stay free.

The covariance normalization is fixed too. Let $\mathsf{F}_g^{(0)}$ denote the center-integrated covariance in the oriented convention of Eq. (3). Define

$$E(s) = \operatorname{diag}(e^{a_\psi s}, e^{a_b s}), \qquad \widetilde{\mathsf{F}}_g(s,s') = E(s)\mathsf{F}_g^{(0)}(s,s')E(s').$$

Then

$$\boxed{\mathsf{S}\widetilde{\mathsf{F}}_{g,p}\mathsf{S} = \sigma\ell_\psi \mathsf{G}_{g,p}, \qquad \ell_\psi = \frac{2\,4^{-\nu}}{J b_\psi^{q-2}}.} \tag{12}$$

For example, the continued fermion rail and its cylinder coefficient are

$$r_\psi = -\frac{2b_\psi^2 \Gamma(\nu)^2 \cos^2(\pi\delta/2)}{\Gamma(2\nu)\cos(\pi\delta)}, \qquad c_\psi = \frac{\Gamma(\delta-1/2)}{2\sqrt{\pi}\Gamma(\delta)}, \qquad r_\psi/c_\psi = \ell_\psi.$$

The boson rail obeys $r_b/c_b = -4\delta^2\ell_\psi$, where $c_b = \Gamma(\delta+1/2)/(2\sqrt{\pi}\Gamma(1+\delta))$. The opposite parity of the two rails supplies the remaining sign. Hence Eq. (12) includes the zero rung, not just the interacting denominator.

These rail integrals require a common homogeneous-distribution continuation: the separate fermion rail has an infrared divergence and the boson rail a short-distance divergence. The identities are not claims of absolute convergence at the physical exponents.

3. The supersymmetric branches

Gamma recurrence gives a useful relation between the two radial responses:

$$R_{\nu,p}(\omega)R_{\delta,1-p}(\omega) = \frac{\mu^2 - \omega^2}{4\delta\nu}. \tag{13}$$

Writing $D_\psi = -[2\nu R_{\nu,p}]^{-1}$, Eq. (9) becomes

$$\widehat{\mathsf{k}}_p = \sigma L_q D_\psi \begin{pmatrix} 2\mu & 2\delta \\ (\omega^2 - \mu^2)/(2\delta) & 0 \end{pmatrix}. \tag{14}$$

Its eigenvalues are $\sigma L_q D_\psi(\mu \pm \omega)$, with eigenvectors

$$u_\pm = \begin{pmatrix} 1 \\ (\pm\omega - \mu)/(2\delta) \end{pmatrix}.$$

Because the eigenvectors depend on frequency, a constant rotation does *not* turn the problem into two independent scalar Robin conditions. At $\omega = 0$, use the original matrix inverse rather than the degenerate eigenvectors.

Let H denote a *superprimary dimension*. Define

$$\begin{aligned} C_\delta &= \frac{\nu\Gamma(\nu)}{\pi\delta\Gamma(\delta)}, \\ k_A(H) &= -C_\delta\Gamma(\delta - H)\Gamma(\delta + H)[\sin(\pi\delta) - \sin(\pi H)], \\ k_S(H) &= -C_\delta\Gamma(\delta - H)\Gamma(\delta + H)[\sin(\pi\delta) + \sin(\pi H)]. \end{aligned} \tag{15}$$

The signs are fixed by the component kernels and the checks $k_A(1) = k_S(1) = 1$, $k_A(0) = k_S(0) = -(q-1)$. The neutral component spectra satisfy

$$\operatorname{spec}\mathsf{k}_1(h) = \{k_A(h), k_S(h-1)\}, \qquad \operatorname{spec}\mathsf{k}_0(h) = \{k_S(h), k_A(h-1)\}. \tag{16}$$

The charged component kernels satisfy $k^{a,d}(h_f) = k_A(h_f - 1/2)$ and $k^{s,d}(h_f) = k_S(h_f - 1/2)$, in the notation of Ref. [2]. Thus a generic root $gk_\alpha(H) = 1$ organizes the dimensions

$$(H,\ H + \tfrac{1}{2},\ H + \tfrac{1}{2},\ H + 1). \tag{17}$$

For $q = 3$, $g = 1$, the first nonexceptional superprimary roots are $H_S = 2.8211441710\ldots$ and $H_A = 3.0659046429\ldots$.

The relative component weights are also retained. At a simple root of one matrix eigenvalue k_α, with the other k_β distinct, let $P_\alpha = (\widehat{\mathsf{k}} - k_\beta I)/(k_\alpha - k_\beta)$. Then

$$\operatorname*{Res}_{h=h_n}[(I - g\widehat{\mathsf{k}})^{-1}\widehat{\mathsf{k}}] = -\frac{P_\alpha(h_n)}{g^2 k'_\alpha(h_n)}. \tag{18}$$

For orientation, at $q = 3$ the two masses and the mixing matrix are simply

$$m_\psi^2 = \frac{7}{36}, \qquad m_b^2 = -\frac{5}{36}, \qquad \mathsf{M}_p = \frac{\sigma 4^{1/3}}{3}\begin{pmatrix} 1 & 2 \\ 2 & 0 \end{pmatrix}.$$

4. An explicit Dirac response for the charged channels

The charged ladder can be represented by a Dirac field of mass $m_F = \mu$. This is more than a mass-pattern guess. In the Lorentzian strip, rescale the physical spinor as $\Psi = \zeta^{1/2}\psi$ and choose

$$H_D = J_2\partial_\theta + \mu\sec\theta\,\sigma_1, \qquad J_2 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad \sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

For $H_D\psi=\Omega\psi$, rotate $\psi=e^{(\pi/4-\theta/2)J_2}(u,v)^T$ and set $\omega=\Omega-1/2$. The equation becomes

$$(\partial_\theta+\mu\tan\theta)u=(\omega+\mu)v,\qquad(-\partial_\theta+\mu\tan\theta)v=(\omega-\mu)u. \tag{19}$$

Squaring gives the two scalar radial equations with masses $\mu(\mu-1)$ and $\mu(\mu+1)$. If $v=\Phi_\psi$ has parity p, then $u=\Phi_b$ has parity $1-p$.

At the right endpoint, Eq. (19) gives

$$A_b=\frac{2\nu}{\omega-\mu}B_\psi,\qquad B_b=-\frac{\omega+\mu}{2\delta}A_\psi. \tag{20}$$

The physical spinor has slow coefficient $a_F=A_\psi$, of weight δ, and fast coefficient $b_F=A_b$, of weight ν. Its response is therefore

$$\boxed{R_{F,p}(\Omega)=\frac{b_F}{a_F}=\frac{2\nu R_{\nu,p}(\Omega-1/2)}{\Omega-\nu}.} \tag{21}$$

The boundary form is $\bar a_{F,1}b_{F,2}-\bar b_{F,1}a_{F,2}$, with coefficient one in the normalized parity basis. With source convention $j_F=b_F$, the free response is R_F^{-1}.

Using the same branch functions as above,

$$\boxed{k_\alpha(\Omega)=\frac{\sigma L_q}{R_{F,p}(\Omega)},\qquad(p,\alpha)=(1,A)\text{ or }(0,S).} \tag{22}$$

Here $\Omega=H=h_f-1/2$. The constant boundary condition and its resolvent are

$$b_F=g\sigma L_q a_F,\qquad G^F_{g,p}=(R_{F,p}-g\sigma L_q)^{-1},\qquad\frac{k_\alpha}{1-gk_\alpha}=\sigma L_q G^F_{g,p}. \tag{23}$$

Equation (20) maps this boundary condition to Eq. (11). A second rotation, $e^{(\pi/4+\theta/2)J_2}$, gives the top scalar at frequency $\Omega+1/2$ and opposite parity; equivalently,

$$R_{F,p}(\Omega)=\frac{2\nu R_{\nu,1-p}(\Omega+1/2)}{\Omega+1/2+\mu}.$$

These are maps between solutions and their boundary data. They do not, by themselves, define supersymmetry transformations of a complete bulk action.

5. From cylinder modes to the four-point function

Choose the chiral–antichiral cross ratio z with $0<z<1$. It is related to the usual component ratio by $\chi=z/(z-1)$. Fix the ordering sign and suppress the common large-N prefactor so that the normalized zero rung is $\mathcal{F}_0(z)=z^\delta$. In this ordering the normalized component fermion covariance is $-\mathcal{F}$.

Define the normalized superblock [3]

$$\varphi_H(z)=\frac{\Gamma(H)^2}{\Gamma(2H)}z^H\,{}_2F_1(H,H;2H+1;z). \tag{24}$$

If $F_H^{\rm bos}=\Gamma(H)^2z^H{}_2F_1(H,H;2H;z)/\Gamma(2H)$, a contiguous identity gives $\varphi_H=F_H^{\rm bos}-F_{H+1}^{\rm bos}$.

The ordinary Abel map in your construction has $\mathcal{A}e^{-(H-1/2)t}=F_H^{\rm bos}/2$. It follows that

$$\boxed{\mathcal{T}_2 f=4\mathcal{A}[\sinh(t/2)f(t)],\qquad\mathcal{T}_2e^{-Ht}=\varphi_H(z).} \tag{25}$$

Explicitly, with $\eta=\operatorname{arcosh}(2/z-1)$,

$$(\mathcal{T}_2f)(z)=2\sqrt{2}\int_\eta^\infty\frac{\sinh(t/2)f(t)}{\sqrt{\cosh t-\cosh\eta}}\,\mathrm{d}t.$$

The starting domain is $\operatorname{Re}H>0$. The super-Casimir is $\mathcal{C}_2=z^2(1-z)\partial_z^2+z(1-z)\partial_z$ and satisfies $\mathcal{C}_2\mathcal{T}_2=\mathcal{T}_2\partial_t^2$ on this mode span. Thus its eigenvalue is H^2 and its shadow is $-H$, not the component shadow $1-h$.

5.1. The normalized density and contour

The block map alone does not give the four-point weights. The component seed overlaps fix

$$\mathcal{N}_\delta = \frac{\delta}{2\nu\tan(\pi\delta/2)}, \qquad D_\delta = \frac{2\mathcal{N}_\delta}{\pi},$$
$$r_g(H) = \mathcal{N}_\delta\left[\cot\frac{\pi H}{2}\frac{k_A(H)}{1-gk_A(H)} - \tan\frac{\pi H}{2}\frac{k_S(H)}{1-gk_S(H)}\right]. \tag{26}$$

Let C_c run upward along $\operatorname{Re} H = c$, $0 < c < \delta$, avoiding dynamical poles. With contour deformations and their residues carried along under continuation,

$$\boxed{\mathcal{F}_g(z) = \int_{C_c}\frac{\mathrm{d}H}{2\pi i}\, r_g(H)\varphi_H(z) + D_\delta\sum_{n=1}^{\infty}\frac{k_A(n)}{1-gk_A(n)}\varphi_n(z).} \tag{27}$$

At integer $n > 0$, $k_A(n) = k_S(n)$. The explicit integer sum cancels the kinematic poles when the contour is closed to the right. It is not an extra tower of exchanged operators. For a nonexceptional root $gk_\alpha(H) = 1$, the remaining block coefficients are

$$c_A(H) = \frac{\mathcal{N}_\delta\cot(\pi H/2)}{g^2k_A'(H)}, \qquad c_S(H) = -\frac{\mathcal{N}_\delta\tan(\pi H/2)}{g^2k_S'(H)}. \tag{28}$$

One useful normalization check is the exact seed identity

$$z^\delta = \sum_{n=0}^{\infty}(-1)^n\frac{\Gamma(2\delta+n)}{\Gamma(\delta)^2 n!}\,\varphi_{\delta+n}(z). \tag{29}$$

It checks all descendant coefficients, not only the leading power. The block sums converge on compact subintervals of $0 < z < 1$, away from pole collisions. Along the vertical contour the full integrand is $O(|\operatorname{Im} H|^{4\delta-5/2})$ and is integrable for $\delta \le 1/3$. This is a prescribed reconstruction, not an inverse on arbitrary center-integrated data.

6. Exceptional terms and an independent check

The origin fixes a constant and a logarithm. Since $\varphi_H = 2/H + 2\log z + O(H)$, the contour must specify which side of $H = 0$ it passes. Put

$$k_0 = -\nu/\delta, \qquad a_0 = k_A'(0) = \frac{\pi\nu}{\delta\sin(\pi\delta)}.$$

The relevant residue is

$$\operatorname*{Res}_{H=0}[r_g(H)\varphi_H(z)] = \frac{4\mathcal{N}_\delta}{\pi}\left[\frac{k_0}{1-gk_0}\log z + \frac{a_0}{(1-gk_0)^2}\right]. \tag{30}$$

The seed in Eq. (29) selects the contour to its right. Moving the upward contour to the left changes the integral by minus this residue.

The $H=1$ multiplet needs a finite-part prescription. At $g=1$, the enhanced S-branch term belongs to the component dimensions $(1, 3/2, 3/2, 2)$. In this particular seed and ordering, the A-branch weight instead vanishes as its root approaches $H=1$. Define

$$\begin{aligned} b_1 &= k_S'(1) = \frac{1-2\delta}{\delta\nu} - \frac{\pi}{\sin(\pi\delta)}, \\ b_2 &= k_S''(1) = b_1^2 + \psi_1(\delta) - \psi_1(\nu) + \nu^{-2} - \delta^{-2}, \end{aligned} \tag{31}$$

where ψ_1 is the trigamma function. A definite subtraction is

$$\mathcal{F}_{\text{sub}}(z) = \lim_{g\to 1^-} \left[\mathcal{F}_g(z) - \frac{D_\delta}{1-g}\varphi_1(z) \right]. \tag{32}$$

Because the pole moves as $H(g) = 1 + (1-g)/b_1 + O((1-g)^2)$, the finite contribution of this multiplet is

$$\boxed{\mathcal{F}_{\text{sub},H=1}(z) = \frac{D_\delta}{b_1}\, \partial_H \varphi_H(z)|_{H=1} + D_\delta \left(1 - \frac{b_2}{2b_1^2}\right) \varphi_1(z).} \tag{33}$$

The full finite part includes the other roots. Equivalently, set $g=1$ in the contour integral and start the explicit integer sum at $n=2$. Deleting the entire exceptional multiplet would lose the derivative block.

At $q=3$, the two coefficients in Eq. (33) are $-0.129565995703\ldots$ and $0.129254113198\ldots$. This is a specified *conformal finite part*, not the finite-coupling super-Schwarzian [4]. Its physical coefficient requires the microscopic lifting of the soft modes. A different finite subtraction can also change the coefficient of φ_1.

6.1. A direct first-rung test

Writing $\mathcal{F}_g = z^\delta + g\mathcal{F}^{[1]} + \cdots$, the original component kernel gives, after one analytically continued convolution,

$$\mathcal{F}^{[1]}(z) = P_\delta C_{\text{conv}} z^\delta \int_{\mathbb{R}} \mathrm{d}v\, \frac{\operatorname{sgn} v\, \operatorname{sgn}(z-v)\, \operatorname{sgn}(v-1)}{|v|^\nu |z-v|^\delta |v-1|^\delta}, \tag{34}$$

where

$$P_\delta = \frac{\nu(1-2\delta)}{2\pi\delta} \tan\frac{\pi\delta}{2}, \qquad C_{\text{conv}} = 2B(\nu, 2\delta-1) - B(\nu,\nu).$$

The remaining integral is absolutely convergent. It uses no conformal blocks or bulk response. It agrees with ∂_g of Eq. (27) at $g=0$; for example,

q	z	$\mathcal{F}^{[1]}(z)$, from both calculations
3	0.2	-0.533994288253294417
3	0.6	0.130574305406873014
5	0.4	-0.371415562926934307

The recorded nine tests at $q=3,5,7$ have maximum discrepancy below 8×10^{-42} at 45-digit working precision, with 60 block terms. Separate contour and root-sum calculations also agree at about 20 digits. These are numerical cross-checks, not certified error bounds or independent external review.

7. What is supersymmetric, and what is not

The component spectrum, relative weights, and reconstructed superblocks have the required CFT supersymmetric organization. For the chiral–antichiral configuration, restoring the super cross ratio and external propagators generates the partner components [3]. This does not make the auxiliary boundary action invariant under the full AdS superalgebra.

There is an explicit necessary-condition test. The standard AdS raising generator acts on a boundary coefficient of weight Δ as

$$E_+^{(\Delta)} = e^{-s}(\partial_s - \Delta).$$

It changes strip parity, while $\mathsf{M}_{1-p} = -\mathsf{M}_p$. For a source-free mode $X = e^{-\omega s}X_0$ satisfying $j_0 = g\mathsf{M}_p X$, use $\Delta_X = \mathrm{diag}(-\mu, 1-\mu)$ and $\Delta_{j_0} = \mathrm{diag}(1+\mu, \mu)$. The second component of the transformed boundary mismatch is

$$\boxed{(j_0' - g\mathsf{M}_{1-p}X')_2 = -2g\omega(\mathsf{M}_p)_{21}(X_0)_1 e^{-(\omega+1)s}.} \tag{35}$$

It is generically nonzero. Thus this fixed boundary domain does not preserve the full standard AdS isometry algebra, and cannot preserve all four standard AdS $\mathcal{N} = 2$ supercharges that close into it. A smaller symmetry or a modified boundary theory is not ruled out. Checking the boundary domain, rather than the mass pattern alone, is essential [5].

The result is a four-point representation. The construction gives fixed-mass scalar and Dirac responses with normalized ladder resummation, followed by a prescribed superconformal reconstruction. It does not supply cubic bulk interactions, an $\mathcal{N} = 2$ six-point calculation, a positive auxiliary vacuum in every continued sector, or a complete gravitational dual. The same boundary conditions cannot be called a fully AdS-supersymmetric completion without changing the setup.

Conventions and calculation record

The component input is from Ref. [2]. The matrix and Dirac factorizations, reconstruction density, and finite part are the derived extension, not claims in the student's draft. The accompanying calculation records are `n2_susy_robin_extension.md` and `n2_completion_supplement.md`. No priority claim is made.

Two printed signs in Ref. [3] need care. Equation (15) uses the component-normalized sign $k_A(1) = k_S(1) = 1$, opposite to the literal overall signs in its Eqs. (4.48)–(4.49). Also its definition $\xi_H = H(\Psi_H^A - \Psi_{H+1}^S)$ and Appendix B imply

$$\xi_H = H\big[(1 + \sec \pi H)\varphi_H - (1 - \sec \pi H)\varphi_{-H}\big],$$

with a minus before the shadow term, unlike its Eq. (4.39). These algebraic discrepancies are not an official erratum. The component overlaps and complete seed independently fix the convention in Eq. (27).

References

[1] K. H. Thong, *Holography with a single bulk field in AdS_{d+1}*, supplied draft, especially Eqs. (3), (11)–(13) and Appendices A–C.

[2] C. Peng, M. Spradlin and A. Volovich, *Correlators in the $\mathcal{N} = 2$ supersymmetric SYK model*, arXiv:1706.06078v2. Component kernels: Eqs. (2.20)–(2.35); four-point overlaps: Sec. 4.

[3] K. Bulycheva, *$\mathcal{N} = 2$ SYK model in the superspace formalism*, arXiv:1801.09006v2. Superblocks, spectral measure and Appendix B; see the convention note above.

[4] W. Fu, D. Gaiotto, J. Maldacena and S. Sachdev, *Supersymmetric SYK models*, arXiv:1610.08917.

[5] D. H. Correa, V. I. Giraldo-Rivera and G. A. Silva, *Supersymmetric mixed boundary conditions in AdS_2 and $DCFT_1$ marginal deformations*, arXiv:1910.04225. Background on spinor boundary data and boundary supersymmetry.